

CyberSafe

An AI-Powered Cybersecurity Assistant for Everyone

Product Requirements Document (PRD)

Version: 1.0

Status: Draft

Owner: Product Team

Our Vision

"A safer digital world for everyone." To make cybersecurity simple, accessible and actionable for every internet user.

Our Mission

To empower individuals with easy-to-use cybersecurity tools that detect threats, explain risks in simple language, and guide them with clear actions.

Product Tagline

Upload • Analyze • Understand • Stay Safe

Our Target Users

- General internet users
- Students
- Professionals
- Small businesses
- Anyone who wants to stay safe online

Key Benefits

- Easy to use (no technical knowledge required)
- Covers multiple types of cyber threats
- Provides clear and actionable guidance
- Helps prevent real-world cyber incidents
- Builds cybersecurity awareness

0. Table of Contents

1. Product Overview
2. User Input Module
3. User Preferences Module
4. Platform Architecture (End-to-End Pipeline)
 - 4.1 Input Processor
 - 4.2 Security Engines
 - 4.3 Risk Engine
 - 4.4 LLM Explanation Layer
 - 4.5 Action Plan
5. Analysis Output (Example Specification)
6. Key Threat Coverage
7. Target Users, Mission & Key Benefits
8. Future Scope

1. Product Overview

CyberSafe is an AI-powered cybersecurity assistant that lets any user submit suspicious digital content (a screenshot, pasted text, a URL, or a file) and receive a clear, risk-scored, and actionable explanation of whether that content is dangerous. The product converts technical security analysis into plain-language guidance, tailored to the user's stated technical level, so that non-experts can understand and act on real threats.

Core Product Flow: Upload → Analyze → Understand → Stay Safe

2. User Input Module

Users can submit any suspicious content they receive or encounter online, through four supported input types:

Screenshot Upload images of emails, messages, websites, pop-ups, etc.	Text Paste suspicious emails, messages, or content directly.
URL Enter a website link to analyze.	File (Optional) Upload documents (e.g. PDF, executables).

3. User Preferences Module

The explanation output is tailored to each user's needs, based on three preference inputs collected before or alongside analysis:

1. Who are you?

Dropdown selection, e.g. Student / Professional / Personal User.

2. How technical should the explanation be?

- **Simple** (Non-technical)
- **Detailed** (Balanced)
- **Technical** (Advanced)

3. What do you want to focus on? (Optional)

- Emails & Messages
- Websites & Links
- Account Security
- Device Security
- Network Security
- Everything

4. Platform Architecture (End-to-End Pipeline)

The platform takes a user's input and processes it through a five-stage pipeline to produce actionable security advice, as follows:

USER INPUT

Screenshot / Text / URL / File



INPUT PROCESSOR

Extract and prepare data for analysis



SECURITY ENGINES

Detect threats using multiple approaches



RISK ENGINE

Combine results to determine overall risk



LLM EXPLANATION LAYER

Convert technical analysis into user-friendly insights



ACTION PLAN

Provide clear, actionable steps



USER

Informed, safer, and better prepared

4.1 Input Processor

Extracts and prepares raw data from the user's submission for downstream analysis.

OCR For images — extracts text from screenshots.	URL Parser Extracts domain, SSL, and metadata from links.	Text Parser Analyzes email / message content.
--	---	---

4.2 Security Engines

Detects threats in the processed data using multiple, complementary approaches.

Rules Engine Signature & pattern matching.	ML Model Classification & anomaly detection.	Security Checks Domain, SSL, sender, attachments, etc.
--	--	--

4.3 Risk Engine

Combines the outputs of the Security Engines to determine the overall risk verdict.

Threat Detection Type of threat.	Severity Assessment Low / Medium / High.	Evidence Collection Reasons & indicators.
--	--	---

4.4 LLM Explanation Layer

Converts the technical risk analysis into user-friendly insights, matching the user's selected technical preference (Section 3):

Simple Easy to understand.	Detailed Balanced explanation.	Technical Advanced details.
--------------------------------------	--	---------------------------------------

4.5 Action Plan

Provides the user with clear, actionable next steps, structured around three questions:

What happened? Describes the detected event.	Why is it risky? Explains the danger.	What should you do? Concrete next steps.
--	---	--

5. Analysis Output (Example Specification)

The output below is the reference example the product must be able to reproduce, shown here for the 'Simple' explanation style:

▲ HIGH RISK Phishing / Social Engineering	Confidence: 94%
---	------------------------

Explanation Style Tabs: **Simple** | Detailed | Technical (user-selectable; Simple shown below)

Simple Explanation

This message is probably a scam. The sender is trying to make you click a link and give away your personal information like your password or OTP.

Why is it suspicious?

- Urgent language
- Suspicious link (not from official domain)
- Requests confidential information
- Sender identity appears fake

What could happen?

Your personal information could be stolen, leading to account compromise or financial loss.

What should you do?

1. Do not click the link.
2. Do not share your password, OTP or personal details.
3. Verify through the official website or contact the organization directly.
4. If you already entered your details, change your password immediately.

6. Key Threat Coverage

The product must detect and explain the following eight threat categories:

Phishing Emails, Messages	Malware Attachments, Files	Malicious URLs Websites, Links	Social Engineering SCAM detection
-------------------------------------	--------------------------------------	--	---

Account Security Passwords, MFA	Device Security OS Configuration	Network Threats MITM, Port Scan, etc.	Web Security SQLi, XSS, SSL/TLS
---	--	---	---

7. Target Users, Mission & Key Benefits

Mission: To empower individuals with easy-to-use cybersecurity tools that detect threats, explain risks in simple language, and guide them with clear actions.

Target Users:

- General internet users
- Students
- Professionals
- Small businesses
- Anyone who wants to stay safe online

Key Benefits:

- Easy to use (no technical knowledge required)
- Covers multiple types of cyber threats
- Provides clear and actionable guidance
- Helps prevent real-world cyber incidents
- Builds cybersecurity awareness